

“Hey, Over Here!”: Comparing App, Voice, and Voice-with-Visual-Feedback for Summoning an In-Office Service Robot

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Abstract

Summoning an in-office service robot today usually means opening an app and tapping through menus. We ask whether *summoning by voice*—saying “over here”—can replace that, and whether adding lightweight *visual feedback* helps people rely on the robot appropriately. The robot estimates the caller’s direction by sound (sound-source localization, SSL) and maps it to a named workstation. Because a physical robot could not be arranged in time, we rebuilt the scenario as a controlled, first-person 3D browser simulation with spatially-positioned audio, and ran a within-subjects study comparing three ways to summon—app, voice only, and voice with visual feedback—across forced correct and mislocalized outcomes plus a multi-user (fairness) scenario. We measure interaction friction, cognitive load, calibrated trust, awareness, and perceived fairness, all with standard HCI instruments. A pilot shows voice substantially lowers interaction friction; the trust and load effects, and a participant-surfaced *social cost of speaking aloud in shared spaces*, motivate context-adaptive interaction as the next step.

Keywords

voice user interfaces, human-robot interaction, sound-source localization, calibrated trust, cognitive load, NASA-TLX, social acceptability, simulation study

1 Introduction

In shared offices and labs, small mobile robots increasingly handle errands such as collecting desk waste. Yet *summoning* such a robot is still clumsy: the user unlocks a phone, opens an app, taps through a menu to indicate a destination, and confirms—several explicit actions that interrupt focused work. This friction is at odds with how people naturally request help from a colleague: they simply call out across the room.

This project (course topic P3) asks whether a service robot can be summoned the same way—by *voice*. The user says “over here”; a microphone array estimates the direction of the voice via sound-source localization (SSL); a semantic map resolves that direction to a *named* workstation; and the robot navigates over. Voice promises lower interaction cost, but it raises two further questions central to human-robot interaction (HRI). First, SSL is imperfect—adjacent desks are acoustically confusable—so the robot will sometimes go to the wrong person; does the user notice, and do they adjust their trust appropriately? We frame this as *calibrated trust* [10]. Second, when several people call at once, does the robot handle them in a way that feels fair?

We originally planned to deploy on a physical robot, but the timeline did not permit it; on the instructor’s advice we rebuilt the scenario as a controlled simulation. This is not merely a fallback: a simulation lets us *force* the rare-but-critical mislocalization events on demand and present every participant the identical stimulus, which is necessary to study calibrated trust cleanly—and it follows the Wizard-of-Oz tradition of simulating not-yet-built or not-under-study components while keeping the studied interaction faithful [2].

Our contributions are: (1) a first-person 3D simulation of voice-summoned office robotics with spatially-positioned audio and a desk-mounted feedback light; (2) a within-subjects study design that isolates interaction friction, calibrated trust, cognitive load, awareness, and fairness with standard instruments; and (3) pilot findings and a participant-surfaced design insight (the social cost of voice in shared spaces) that motivate context-adaptive interaction.

2 Related Work

Voice interaction and its social cost. Voice lowers input effort, but speaking aloud is not always socially appropriate: users report reluctance to use voice assistants in shared or public spaces because of disturbance and self-consciousness [4]. This tension—convenience versus social acceptability—is central to our findings.

Sound-source localization. Microphone-array SSL estimates a bearing from inter-microphone time differences; open frameworks such as ODAS provide real-time localization and tracking [6]. Its characteristic failure—confusing nearby sources—is exactly the error we model.

Robot legibility and awareness. For users to anticipate a robot’s behavior, its motion and intent must be *legible* [3]; more broadly, supporting the operator’s situation awareness is a long-standing goal [5]. Our desk light and on-screen cues are designed to make the robot’s intent legible to a seated user who cannot see the whole room.

Trust and calibrated trust. Trust in automation is well-characterized: it should be *calibrated* to the system’s actual reliability—high when the system is right, low when it is wrong [10]—and can be measured with validated scales [9]. Our central hypothesis concerns whether feedback improves this calibration when the robot mislocates the caller.

Workload. The NASA-TLX [7] is the standard subjective workload instrument; we adapt its *Mental Demand* subscale as a single, per-trial proxy for cognitive load.

Fairness. Perceived fairness lacks a single canonical HRI scale; we draw on the organizational-justice tradition [1] and treat fairness as an emerging measure for the multi-user case.

3 The Hey, Over Here! System

3.1 Scenario and pipeline

A user seated at a named workstation summons the robot. A voice summon proceeds in four stages: (1) the user says “over here”; (2) SSL estimates a bearing; (3) a *semantic map* resolves the bearing to a named desk—crucially, the map knows “that is Finn’s desk,” turning a bare angle into an actual destination; (4) the robot navigates over and collects the waste.

3.2 Simulation

Because the physical robot was unavailable, each stage is implemented in a self-contained browser simulation. The interaction layer (how one summons, and what feedback is shown) is kept faithful, while perception and actuation are replaced by software models: SSL is a true bearing plus Gaussian angular noise ($\sigma \approx 11^\circ$) snapped to the nearest desk, reproducing the confuse-adjacent-desks failure; navigation is waypoint steering. Because the pipeline and interaction are unchanged, HCI findings are intended to transfer to the real robot. The live simulation and study tool are publicly available at <https://lyqgwzx.github.io/hey-over-here-p3/>.

A key methodological refinement was moving from a 2D top-down view to a *first-person* 3D office. In the top-down view the whole room—and the robot—is always visible, so even in the “no-feedback” condition a participant can simply watch the robot move; this leaks the robot’s destination to every condition and confounds the feedback manipulation. In first person, seated in a semi-enclosed cubicle, the participant cannot see the whole room: the robot may be behind them or occluded, so the no-feedback condition is genuinely low-information. The robot emits 3D spatial audio (HRTF panning), so an unseen robot can still be localized by ear; we return to this in the Limitations.

3.3 Conditions and feedback

Three summon methods are compared. *App* (baseline): open an app, pick a desk, confirm—several taps, no SSL. *Voice only*: hold a button and say “over here”—one action, but no lights or screen. *Voice + visual*: the same voice summon, plus a light on the user’s own desk that turns on when the robot has locked onto them and flashes faster as it nears (pale yellow far, bright red close); in the multi-user case, an on-screen badge acknowledges the user, states their queue position, and gives an ETA. Crucially, feedback is delivered *to the user’s location*, not via the robot, precisely because a seated user cannot see through walls.

4 Study Design

4.1 Design and scenarios

The study is within-subjects: each participant experiences six guided scenarios—a baseline, a 2x2 of feedback (voice / voice+visual) x localization outcome (correct / mislocalized), and a multi-user fairness scenario:

- (1) **App** baseline (always correct; no SSL).
- (2) **Voice · correct**.
- (3) **Voice · mislocalized** (robot goes to a neighbour).
- (4) **Voice+visual · correct**.
- (5) **Voice+visual · mislocalized**.

Table 1: Measures, constructs, and sources.

Measure	Construct	Source
Taps + time-to-arrival	Interaction efficiency	ISO 9241-11 [8]
“How mentally demanding was this?” (1–5)	Cognitive load	NASA-TLX Mental Demand [7]
“I trusted it to come to the right desk.” (1–5)	Trust / calibrated trust	[9, 10]
“I always knew where the robot was going.” (1–5)	Awareness / legibility	[3, 5]
“It handled everyone fairly.” (1–5)	Perceived fairness	cf. [1]

- (6) **Multi-user fairness**: a colleague is already being served; the user summons and is acknowledged, queued, and given an ETA.

The five single-summon scenarios are randomized; the fairness scenario is last. The localization outcome is *forced* within the voice and voice+visual conditions so that every participant experiences exactly one correct and one mislocalized trial of each—a balanced exposure that makes the within-subjects contrast clean and comparable.

4.2 Measures

After every summon, participants answer the same short, identically-worded items so the methods are comparable on one scale (Table 1). Behavioral measures (taps, time-to-arrival, localization outcome, queue wait) are logged automatically and complement the self-reports.

4.3 Calibration contrast (primary hypothesis)

Calibrated trust is not a single self-report item but a *pattern*: for each participant we compute the drop in confidence from correct to mislocalized trials, Δ_{voice} and Δ_{visual} . The hypothesis is $\Delta_{\text{visual}} > \Delta_{\text{voice}}$: visual feedback should make confidence fall more appropriately when the robot errs, i.e. trust better tracks reality.

4.4 Procedure and data

Participants are run in person on one laptop. Each enters an identifier, completes the six scenarios, and rates each; trials accumulate in the browser (with local persistence against accidental loss) and are exported as a single CSV—one row per summon, recording participant, scenario order (step), condition, localization outcome, behavioral measures, and ratings. Because each row is self-describing, the randomized order does not affect grouping; step additionally permits checking for order effects. Statistical analysis uses within-subjects (paired / repeated-measures) tests, with non-parametric tests (Wilcoxon signed-rank, Friedman) for the 1–5 ordinal items and effect sizes reported alongside.

Table 2: Per-condition means (pilot, $N=5$). Ratings 1–5; *aware* = “I always knew where it was going,” *conf* = confidence it would reach the right desk.

Condition	taps	effort	aware	conf
App (baseline)	3.0	2.8	4.8	4.8
Voice · ok	1.0	2.8	2.8	3.2
Voice · wrong	1.0	2.0	2.4	2.6
V+visual · ok	1.0	1.4	4.4	3.6
V+visual · wrong	1.0	2.0	2.4	2.4
Fairness (multi-user)	1.0	2.0	4.0	4.2

5 Results

We report a pilot of five participants, each completing all six scenarios (30 trials). Given the small sample we report descriptive statistics only, with no significance testing. Crucially, because a mean over five people can be driven by one or two outliers, we also report *how many of the five participants individually showed each effect*—a more honest summary at this scale. Table 2 gives the per-condition means.

Interaction friction (5/5). All five participants took one action per summon by voice or voice+visual versus three for the app—a unanimous, clear reduction. (Time-to-arrival varies with the target desk’s distance, so taps are the primary friction measure.)

Cognitive load (3/5). Voice+visual on correct trials was rated least mentally demanding (1.4/5), below both voice-only and the app (2.8 each). Three of five rated voice+visual easier than voice-only and the other two tied it—none rated it harder. Notably, voice *alone* was no less demanding than the app; the reduction came from the *visual* feedback, not from voice per se.

Awareness (4/5). Voice+visual raised awareness over voice-only on correct trials (4.4 vs. 2.8), and four of five participants individually showed this, indicating the desk light helped people know what the robot was doing.

Calibrated trust (suggestive, 3/5). When the robot went to the wrong desk, confidence fell under both modalities. On average the drop was larger under voice+visual (3.6→2.4) than voice-only (3.2→2.6)—the direction predicted for better-calibrated trust—but this was *not* consistent: only three of five participants showed the larger drop (the mean is inflated by two of them). We therefore treat calibrated trust as at most suggestive at this sample size, and the clearest test of our primary hypothesis for a larger study.

Fairness. The multi-user acknowledge–queue–ETA handling was rated 4.6/5 even though participants genuinely waited behind a colleague. Queue waits ranged from near-zero to about 8 s (mean 4.5 s), yet fairness stayed high and degraded only gently with the wait: the three shortest waits were each rated 5/5, and even the two who waited longest (≈ 8 s) still rated it 4/5. This suggests the on-screen, location-delivered cues convey fairness even when the robot is out of sight and the wait is non-trivial.

6 Discussion

Visual feedback as the increment over audio. Because the robot is audible in every condition, the study measures what visual feedback adds *on top of* sound, not in a vacuum. We therefore

expect its value to appear less as raw awareness (which audio already supplies) and more as *lower cognitive load* and *more certain, better-calibrated* confidence—the reassurance of an unambiguous, glance-able signal, especially in the acoustically hard case where the robot goes to an adjacent desk.

Why queueing feels fair, not surprising. A single robot serving several callers must make someone wait, which could easily feel arbitrary or unfair. Two cues keep the wait transparent and expected. First, while another caller is being served, the user is told *who* the robot is currently helping and *how long* until it reaches them—a live, updating ETA. Second, when the robot does turn to them, their desk light comes on. Because the user is acknowledged and kept informed throughout, and the hand-over is clearly signalled, waiting their turn is predictable rather than opaque—so it does not read as unexpected, out of order, unfair, or uncomfortable. This is consistent with the high fairness rating, and it turns an inherent constraint (one robot, several people) into an acceptable experience.

The social cost of voice (a participant insight). A participant observed that while voice is more convenient than the app, speaking aloud in an office may disturb others. This surfaces a dimension beyond efficiency—the *social acceptability* of voice [4]—that the app does not incur. The preferable modality is thus *context-dependent*: voice clearly wins when one is alone or at home, while a shared, quiet office carries a social cost. This points to *context-adaptive modality*: sensing occupancy or ambient noise and switching between voice and a silent input—convenient when alone, unobtrusive when others are present.

7 Limitations

Audio in every condition. The robot is audible throughout, so a voice-only participant can partly localize it by ear; the honest comparison is audio vs. audio+visual, i.e. the *increment* of visual feedback. We deliberately do not muffle the audio to manufacture an effect. **On-screen vs. co-located.** A screen and desk light are weaker stimuli than a real robot lighting up beside the user, so absolute affect may not transfer. **Simulated SSL.** Localization error is a Gaussian model, not real acoustics. **Pilot scale.** The sample is small and a convenience sample; effects are indicative, not confirmatory.

8 Future Work and Deployment

The immediate step is a full within-subjects sample to test the calibration contrast. Beyond that, Phase 2 ports the validated interaction layer onto the physical Leon robot, replacing the scripted perception and navigation with a real microphone array and ODAS for SSL [6] and SLAM-based navigation, and re-running a small field study to confirm that the effects hold under real embodiment and real error distributions. The participant-surfaced social-cost finding motivates *context-adaptive modality* as a concrete design direction.

9 Conclusion

We compared app, voice, and voice-with-visual-feedback summoning of an in-office service robot in a controlled first-person simulation with spatial audio. Voice lowers the cost of summoning; visual

feedback is designed to support calibrated trust and fairness, delivered to the user’s location rather than via an occluded robot; and the whole pipeline is measured and reproducible. A user insight—that voice carries a social cost in shared spaces—reframes “which is better” as context-dependent and motivates context-adaptive interaction as the path to deployment.

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